

Three-dimensional fiber network structure reinforced composite solid-like electrolyte for lithium metal batteries

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Experimental sections

Synthesis of ZIF-8

Solution A was formed by dissolving 3.92 g of $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ in 100 mL of methanol solution, and solution B was formed by dissolving 8.66 g of 2-methylimidazole (2-MeIm) in another 100 mL of methanol solution. A was quickly poured into solution B, mixed and stirred for 10 min, stood at room temperature for 12 h, and white ZIF-8 nanocrystals were obtained after centrifugation and drying.

Synthesis of Solid-like Electrolyte

Firstly, polyvinylidene fluoride-hexafluoropropylene (PVDF-HFP) solid and ZIF-8 particles were dissolved in a mixed solution of N, N-dimethylformamide (DMF) and acetone (volume ratio of 1:1). The mass fraction of PVDF-HFP was controlled to be 10 wt.%, and the mass fraction of ZIF-8 crystal was 20 wt.%. The PVDF-HFP/ZIF-8 fiber network was obtained by electrospinning. In order to set up the control group, the polymer fibers containing only PVDF-HFP (10 wt.%) were prepared by the same method. Subsequently, PVDF-HFP/ZIF-8 fiber film and pure PVDF-HFP fiber were immersed in Li-ILs (1.0 M LiTFSI dissolved in [EMIM][TFSI]) at room temperature for 12 h, and then immersed in a vacuum oven at 80 °C for 12 h to obtain PZL and PL solid electrolyte films.

Material Characterization

X-ray diffraction (XRD) data were recorded by a Kratos XRD-700 using $\text{Cu K}\alpha$, $\lambda=1.541$, $2\theta=5^\circ\text{-}80^\circ$. The field emission scanning electron microscopy (FESEM) morphology and energy dispersive spectrometer (EDS) mapping were investigated

using FEI Verios G4 scanning electron microscopy with the Thermo NS7 energy dispersive spectrometer. Thermogravimetric analysis (TGA) was carried out in N₂ atmosphere with a heating rate of 10 °C min⁻¹ on a Mettler Toledo TGA STAR system. Differential Scanning Calorimetry (DSC, Mettler–Toledo) tests were performed in air gas at a heating rate of 10 °C min⁻¹.

Electrochemical Performance Tests

Ionic conductivity (IC), linear sweep voltammetry (LSV) and Li⁺ transference number (t_{Li^+}) were collected with a Gamry Interface 1010 electrochemical workstation. For the IC, the SLEs was sandwiched between two stainless steel plates at temperatures from 30 to 80 °C and the frequency range was 0.1 to 10⁵ Hz (SS/SLEs/SS). The IC was calculated by the following formula.

$$\sigma = \frac{L}{RA} \quad \text{Eq.1}$$

Where L represents the thickness of the SLEs, R represents the impedance, and A is the area of the stainless-steel (16 mm disc).

$$\sigma_T = A \exp\left(\frac{-E_a}{RT}\right) \quad \text{Eq.2}$$

Where σ_T is the ionic conductivity at different temperatures, E_a is the activation energy, T is the absolute temperature, and A is the pre-exponential factor. The LSV test was performed by assembling a Li/SLEs/SS cell at a scan rate of 0.5 mV s⁻¹ from 1.0 to 6.0 V to evaluate the electrochemical stability of the SLEs to Li metal. t_{Li^+} was measured by the chronoamperometry and calculated according to following formula.

$$t_{Li^+} = \frac{I_{ss}(\Delta V - I_0 R_0)}{I_0(\Delta V - I_{ss} R_{ss})} \quad \text{Eq.3}$$

Where I_0 is the initial current, I_{ss} is the steady-state current, R_0 is the interface

impedance before polarization, and R_{ss} is the interface impedance after polarization, ΔV is voltage bias (10 mV). LMBs are prepared by using LiFePO_4 (LFP), SLEs, Li metal. The LiFePO_4 /SLEs/Li LMBs were charged and discharged between 2.7-3.8 V at room temperature. Li stripping/plating cycles (the Li symmetrical battery was charged and discharged for 0.5 h at room temperature with a current density of 0.2 mA cm^{-2}) and battery cycling performance were obtained with a LAND battery cycler.

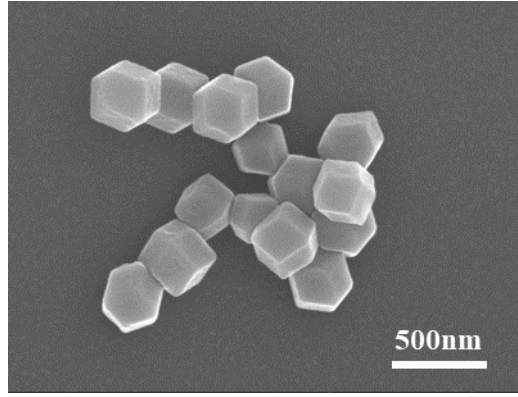


Fig. S1 FESEM image of ZIF-8.

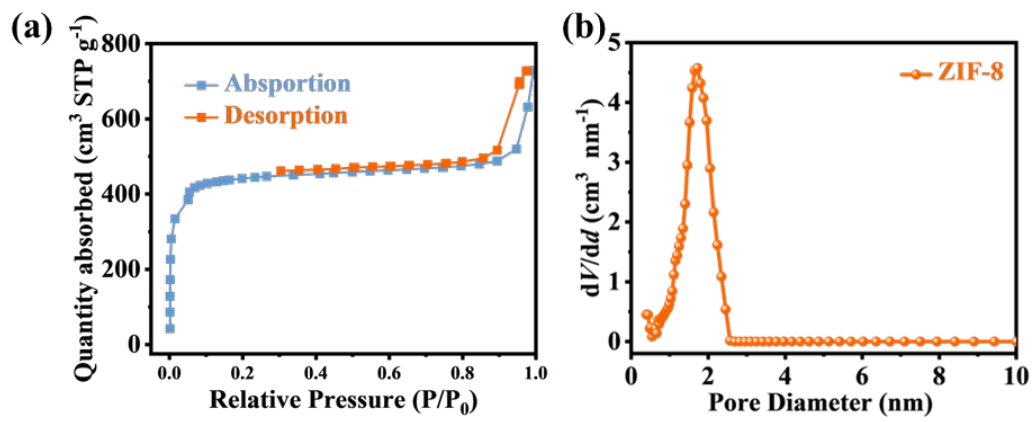


Fig. S2 (a) The N_2 adsorption/desorption curves of the ZIF-8. (b) the corresponding pore size distribution curves.

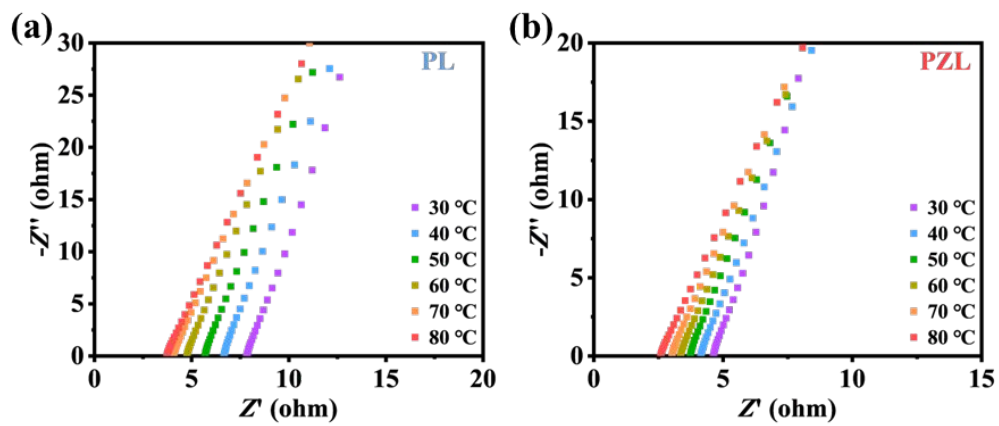


Fig. S3 Nyquist plots of PL and PZL from 30 °C to 80 °C.

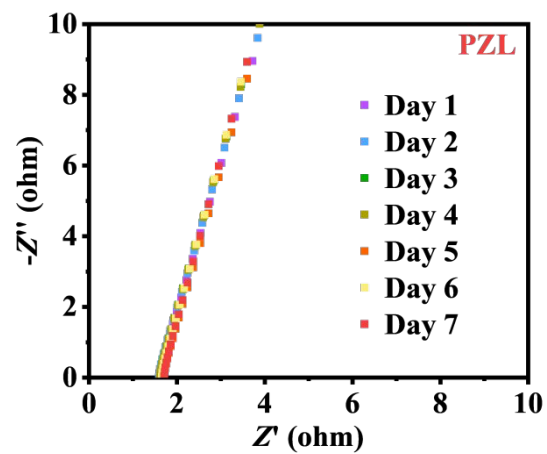


Fig. S4 Nyquist plots of PL and PZL from Day1 to Day7.

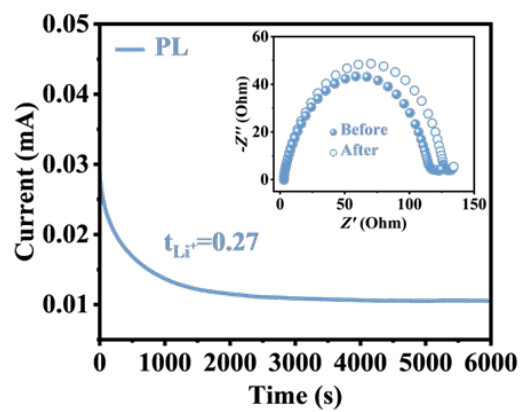


Fig. S5 Li^+ transference number of PL.

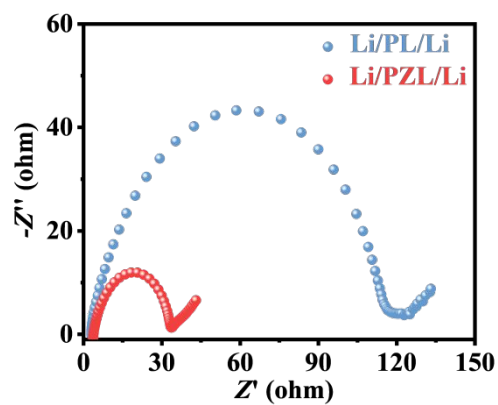


Fig. S6 EIS curves of Li/PL/Li and Li/PZL/Li .

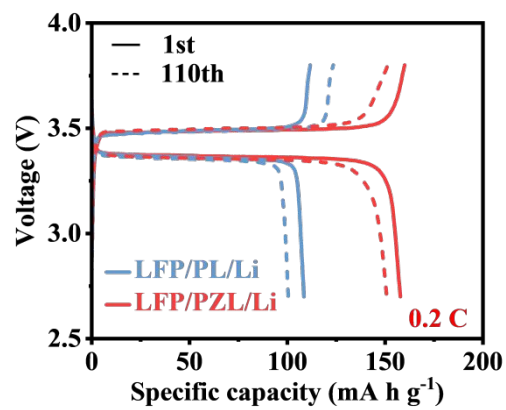


Fig. S7 Charge/Discharge curves before and after 110 cycles.

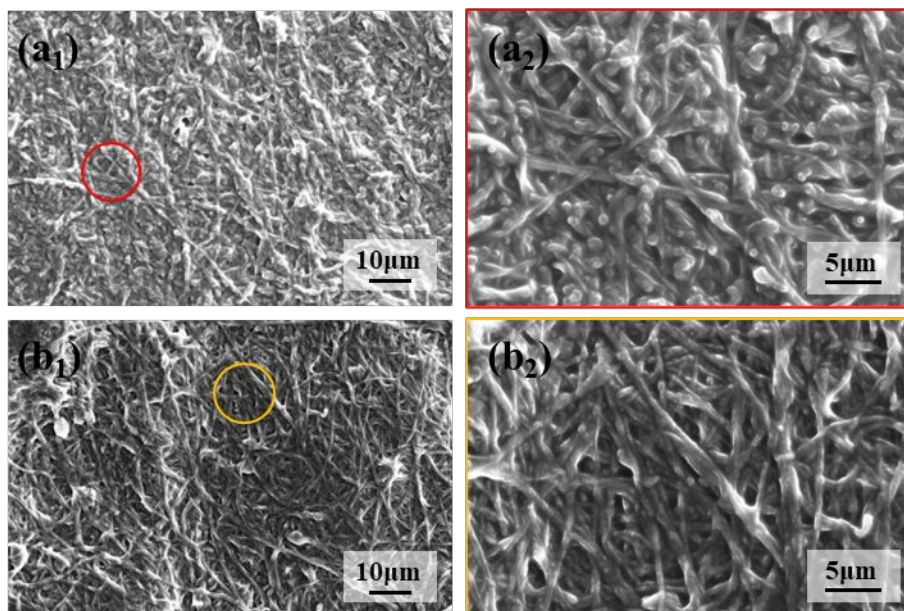


Fig. S8 FESEM images of PZL (a) after 20 cycles, (b) after 100 cycles.